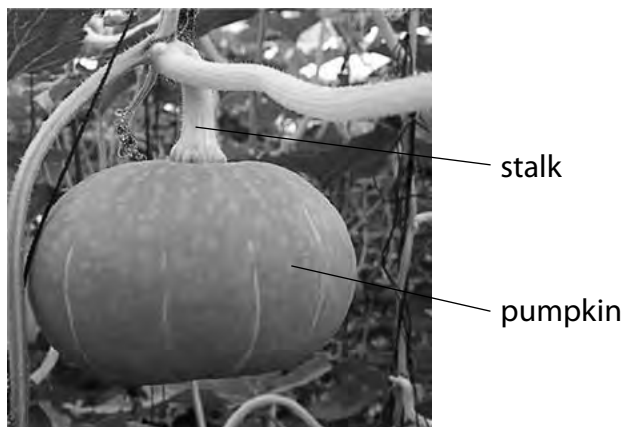


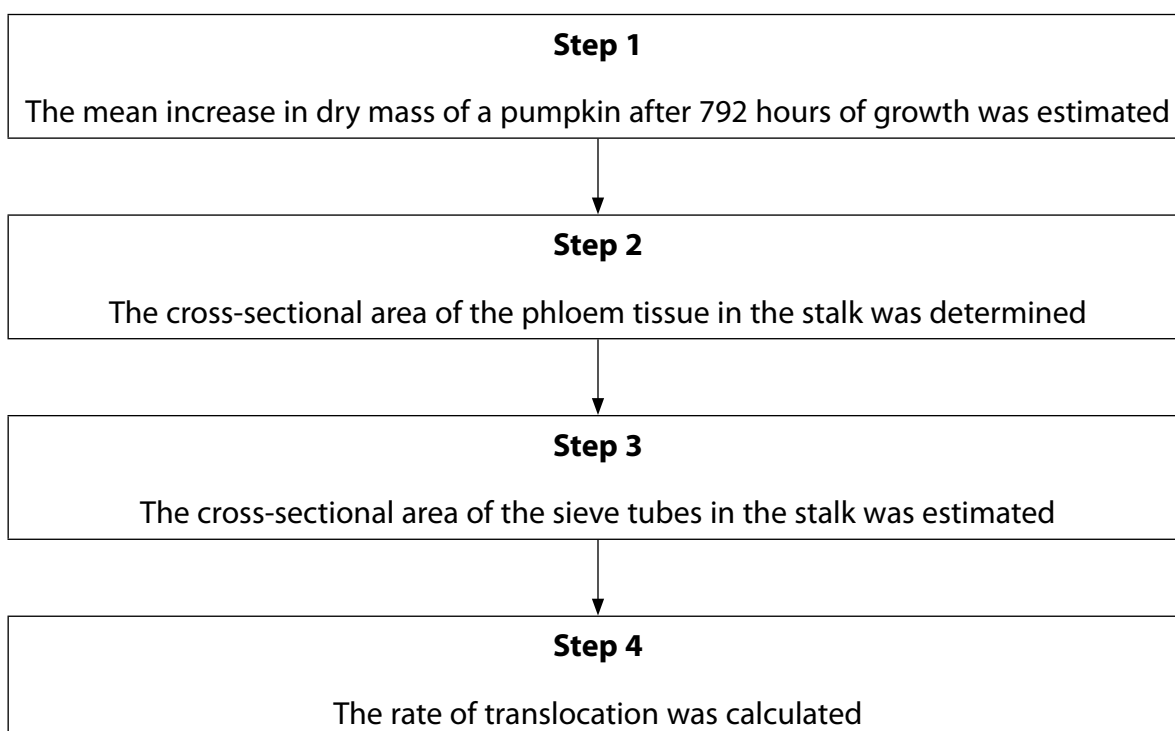
5 The photograph shows a pumpkin.



Source from: https://www.aliexpress.com/price/winter-outdoor-plants_price.html

The scientists Crafts and Lorenz investigated the rate of translocation through the phloem in pumpkins.

The flow chart shows the method used in this investigation.



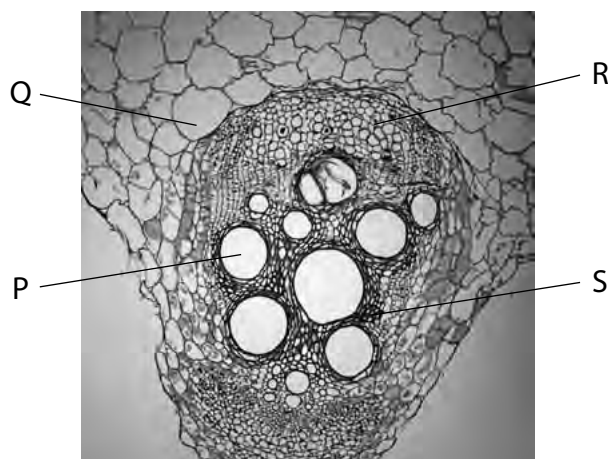
- (a) (i) Suggest how the mean increase in dry mass of a pumpkin could be estimated in **Step 1**.

(3)

- (ii) Explain why Crafts and Lorenz used dry mass in this investigation.

(2)

- (b) (i) The photograph shows a cross-section through part of a stalk, as seen using a light microscope.



Which letter is pointing to the phloem?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

- (ii) Describe a method that could be used to determine the cross-sectional area of the phloem in **Step 2**.

(2)

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- (c) Give a reason why only the cross-sectional area of the sieve tubes, rather than the phloem tissue, was estimated in **Step 3**.

(1)

- (d) What are the units for the rate of translocation calculated in **Step 4**?

(1)

- ☐ **A** $\text{g cm}^{-2} \text{ hr}^{-1}$
- ☐ **B** $\text{g cm}^2 \text{ hr}^{-1}$
- ☐ **C** $\text{g cm}^{-3} \text{ hr}^{-1}$
- ☐ **D** $\text{g cm}^3 \text{ hr}^{-1}$

(Total for Question 5 = 10 marks)